

06:01

- **Pclass**: Passenger class (1 = First, 2 = Second, 3 = Third).
- **Gender**: Gender of the passenger

Expected output

202501050046@mitaoe.ac.in ▾ Support Logout ↗

01:40 🔍 🌙 📝 🔗 —


[Histograma...](#)

Expected output

202501050046@mitaoe.ac.in ▾ Support Logout ↗

55:52 🔍 🌙 📝 🔗 —

rer  BarPlotOf..

Expected output

202501050046@mitaoe.ac.in ▾ Support Logout ↗

20:18 A 🌙 ✎ 🔗 -

rer  BarPlotOf...

Expected output



CODETANTRA

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5.2.5. Bar Plot for Survival by Pclass18:46

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Pclass"** to the chart.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time0.761 sMaximum time0.761 s1 out of 1 shown test case(s) passed

Test case 1761 msDebug

Expected output

Survival by Pclass

CODETANTRA

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5.2.6. Bar Plot for Survival by Embarked33:18

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset. The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title **"Survival by Embarked"** to the chart.
4. Label the x-axis as **'Embarked'** and the y-axis as **'Count'**.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as **'Survived'** and **'Not Survived'**).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time0.613 sMaximum time0.613 s1 out of 1 shown test case(s) passed

Test case 1613 msDebug

Expected output

Survival by Embarked

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5.2.7. Box plot for Age Distribution10:38

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time0.535 sMaximum time0.535 s1 out of 1 shown test case(s) passed

Test case 1535 msDebug

Expected output

CODETANTRA

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5.2.8. Box Plot for Age by Survived17:05

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time0.528 sMaximum time0.528 s1 out of 1 shown test case(s) passed

Test case 1528 msDebug

Expected output

Age by Survival

5.2.9. Box Plot for Fare by Pclass

01:36

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with `plt.suptitle("")`.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time

0.544 s

544.00 ms

Maximum time

0.544 s

544.00 ms

1 out of 1 shown test case(s) passed

Test case 1 544 ms

Debug

Expected output



5.2.10. Scatter Plot for Age vs. Fare

01:36

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time

0.404 s

404.00 ms

Maximum time

0.404 s

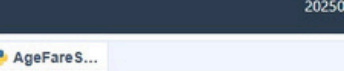
404.00 ms

1 out of 1 shown test case(s) passed

Test case 1 404 ms

Debug

Expected output



5.2.11. Scatter Plot for Age vs. Fare by Survived

08:55

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time

0.519 s

519.00 ms

Maximum time

0.519 s

519.00 ms

1 out of 1 shown test case(s) passed

Test case 1 519 ms

Debug

Expected output

